SECURITY MARKET INDICES

PROBLEMS

1. When creating a security market index, an index provider must first determine the:
   A. Target market.
   B. Appropriate weighting method.
   C. Number of constituent securities.

2. One month after inception, the price return version and total return (+ income (div)) version of a single index (consisting of identical securities and weights) will be equal if:
   A. Market prices have not changed.
   B. Capital gains are offset by capital losses.
   C. The securities do not pay dividends or interest.

22:00

3. An analyst gathers the following information for an equal-weighted index comprised of assets Able, Baker, and Charlie:

Security	Beginning of Period Price (R)		End of Period Price (R)		Total Dividends (R)
Able	10.00	+2	12.00	20%	0.75
Baker	20.00	-1	19.00	- 5%	1.00
Charlie	30.00	0	30.00	0%	2.00

The price return (just cap gains) of the index is:
   A. 1.7%.
   B. 5.0%.
   C. 11.4%.

60 61

∴ $\frac{1}{3}$ (20 - 5%)
= 5%

4. An analyst gathers the following information for a price-weighted index comprised of securities ABC, DEF, and GHI:

Security	Beginning of Period Price (R)		End of Period Price (R)		Total Dividends (R)
ABC	25.00	33%	27.00	-2	1.00
DEF	35.00	47%	25.00	-10	1.50
GHI	15.00	20%	16.00	+1	1.00

The price return of the index is:
   A. -4.6%.
   B. -9.3%.
   C. -3.9%.

75 68

$\frac{7}{75} \times 100$

33% × 8%
47% × 28,57%
20% × 6,67%
≈ -14,73

5. An analyst gathers the following information for a market-capitalization-weighted index comprised of securities MNO, QRS, and XYZ:

Security		Beginning of Period Price (R)		End of Period Price (R)	Dividends Per Share (R)	Shares Outstanding
MNO	23,26%	2,500	× 5k	2,700	100	5,000
QRS	48,84%	3,500	× 7,5k	2,500	150	7,500
XYZ	27,91%	1,500	× 10k	1,600	100	10,000

12,5k m | 13,5k m | + | 500k | 12%
26,25k m | 18,75k m | | 1,125m | -24,29%
15k m | 16k m | | 1m | 13,33%
53,75k m | 48,25m | | 2,625m | -5,35

CG + Div

The total return of the index is:

$\frac{P_1 - P_0 + Div}{P_0}$

$\frac{48{,}25 - 53{,}75 + 2{,}625}{53{,}75}$

A. 1.04%.
B. -5.35%.
C. -10.23%.

6. An analyst gathers the following data for a value-weighted index:

	Beginning of Period			End of Period			
Security	Price (R)	Shares		Price (R)	Shares		
A	20.00	300	6k	22.00	300	6,6k	10%
B	50.00	300	15k	48.00	300	14,4k	-4%
C	26.00	2,000	52k	30.00	2,000	60k	15,38%
			73			81k	

The return on the value-weighted index over the period is:

A. 7.1%.
B. 11.0%.
C. 21.4%.

$\frac{8}{73} \times 100$

$= 10{,}96\%$

7. If the price (no Div) return of an equal-weighted index exceeds that of a market-capitalization-weighted index comprised of the same securities, the *most likely* explanation is:

A. Stock splits.
B. ~~Dividend distributions.~~
C. Outperformance of small-market-capitalization stocks.

8. A float-adjusted market-capitalization-weighted index weights each of its constituent securities by its price and:

A. Its trading volume.
B. The number of its shares outstanding.
C. The number of its shares available to the investing public.

9. Which of the following index weighting methods is *most likely* subject to a value tilt?

A. Equal weighting.
B. Fundamental weighting.
C. Market-capitalization weighting.

10. Which of the following statements regarding sector indices is *most* accurate? Sector indices:

A. Track different economic sectors and cannot be aggregated to represent the equivalent of a broad market index.
B. Provide a means to determine whether an active investment manager is more successful at stock selection or sector allocation.
C. Apply a universally agreed-upon sector classification system to identify the constituent securities of specific economic sectors, such as consumer goods, energy, finance, health care.

11. Which of the following is an example of a style index? An index based on:

A. Geography.
B. Economic sector.
C. Market capitalization.